

Rate Limiter :

Functional requirements:

For my distributed system my rate limiter should decide whether we need to accept/reject the upcoming request ,
rate limiter should be able to handle increased load/increased traffic

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,

CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime
i.e Availability >> Consistency)

Core Entities:

user,
request,
response,
history

API :

```
POST:request/  
{  
    userID,  
    requestID  
}
```

HLD Architecture:

